COMS 6998 - High Performance Machine Learning
Homework Assignment 3

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Due Date: March 25, 2026 | Spring 2026 | Max Points: 110

Instructions:

This lab is intended to be performed **individually**, great care will be taken in verifying that students are authors of their own submission.
This lab has three parts. Part-A on vector addition and matrix multiplication in CUDA, Part-B on CUDA Unified Memory, and Part-C on convolution in CUDA. There is an optional bonus question in Part-C related to using Open AI Triton. Please perform this lab on a GCP GPU instance (setup instructions are at the end of the document) or the Insomnia cluster. As we get closer to the deadline, we might experience GPU availability issues. Start early. You can also use other GPU infrastructure that you might have access to. Please document in your report which setup you used.

The submission template for this assignment is:

Submit a 'UNI.zip' file with the following contents

- Part-A (directory)
  - Makefile
  - vecadd.cu
  - vecaddKernel.h
  - vecaddKernel00.cu
  - vecaddKernel01.cu
  - matmultKernel00.cu
  - matmultKernel01.cu
  - timer.h
  - timer.cu
- Part-B (directory)
  - q1.cpp
  - q2.cu
  - q3.cu
  - q4_with_unified.jpg
  - q4_without_unified.jpg
  - q4.py (the code to plot these results in python or any other language of your choice)
- Part-C (directory)
  - c1.cu

  - c2.cu
  - c3.cu
  - program_output.csv (The program output described in Part-C; can be generated manually)
  - c5.ipynb

- report.pdf (The report with outputs from all the parts and answers to questions wherever requested)

Your report must contain the setup used, screenshots of outputs and execution times from the GCP instance for all programming questions. Paste screenshots for all values of K, threads, and blocks in Part-B Q1, Q2, and Q3. An easy way to time a program is to use the **time** utility in bash. We expect you to maintain academic honesty when submitting your screenshots. Submitting dishonest outputs will invite a strict penalty.

Part-A: CUDA Matrix Operations (40 points)

The purpose of this exercise is for you to learn how to write programs using the CUDA programming interface, how to run such programs using an NVIDIA graphics processor, and how to think about the factors that govern the performance of programs running within the CUDA environment. For this assignment, you will modify two provided CUDA kernels. The first is part of a program that performs vector addition. The second is part of a program that performs matrix multiplication. The provided vector addition program does not coalesce memory accesses. You will modify it to coalesce memory access. You will modify the matrix multiplication program to investigate its performance and how it can be optimized by changing how the task is parallelized. You will create an improved version of the matrix multiplication program and empirically test the time it takes to run. You will analyze the results of your timing tests.

You will turn in all code you have written and used for this assignment, a makefile that will compile your code, and a standard lab report documenting your experiments and results. We need to be able to compile your code by executing "make".

Note: The GCP setup instructions for this part have been provided at the end of this document under Appendix.

Problem-1: Vector add and coalescing memory access

You are provided with `vecadd`, a micro benchmark to determine the effectiveness of coalescing. The provided code is a non-coalescing version, and it is your job to create a coalescing version, and to measure the difference in performance of the two codes.

Here is a set of files provided in `CUDA1.zip` for this problem

- `Makefile`
- `vecadd.cu`

- `vecaddKernel.h`
- `vecaddKernel00.cu`
- `timer.h`
- `timer.cu`

Q1 **5 points**

Compile and run the provided program as `vecadd00` (using the kernel `vecaddKernel00`) and collect data on the time the program takes with the following number of values per thread: 500, 1000, 2000.

Include a short comment in your make file describing what your various programs do, and fully describe and analyse the experiments you performed in your lab report.

Q2 **10 points**

In `vecadd01` you will use a new device kernel, called `vecAddKernel01.cu`, to perform vector addition using coalesced memory reads. Change the `Makefile` to compile a program called `vecadd01` using your kernel rather than the provided kernel. Modify the `Makefile` as appropriate so the clean and tar commands will deal with any files you have added.

Test your new program and compare the time it takes to perform the same work performed by the original. Note and analyse your results, and document your observations in the report.

Problem-2: Shared CUDA Matrix Multiply

For the next part of this assignment, you will use `matmultKernel00` as a basis for another kernel with which you will investigate the performance of matrix multiplication using a GPU and the CUDA programming environment. Your new kernel should be called `matmultKernel01.cu`. Your code files must include internal documentation on the nature of each program, and your `README` file must include notes on the nature of the experiment. You can have intermediate stages of this new kernel, eg without and with unrolling, but we will grade you on your final improved version of the code and its documentation.

Here is a set of provided files in `CUDA1.zip` for this problem:

- `Makefile`
- `matmult.cu`
- `matmultKernel.h`
- `matmultKernel00.cu`
- `timer.h`
- `timer.cu`

In both the questions of this problem (Q3 and Q4 below) you should investigate how each of the following factors influences performance of matrix multiplication in `CUDA`:

1. The size of matrices to be multiplied
2. The size of the block computed by each thread block
3. Any other performance enhancing modifications you find

Q3 **8 points**

First you should time the initial code (with the provided kernel `matmultKernel00`) using square matrices of each of the following sizes: 256, 512, 1024.
When run with a single parameter, the provided code multiplies that parameter by `FOOTPRINT_SIZE` (set to 16 in `matmult00`) and creates square matrices of the resulting size. This was done to avoid nasty padding issues: you always have data blocks perfectly fitting the grid.

Q4 **12 points**

In your new kernel each thread computes four values in the resulting C block rather than one. So now the `FOOTPRINT_SIZE` becomes 32. (Notice that this is taken care of by the `Makefile`.) You will time the execution of this new program with matrices of the sizes listed above to document how your changes affect the performance of the program. Provide the speedup that your new kernel achieves over the provided kernel.

To get good performance, you will need to be sure that the new program coalesces its reads and writes from global memory. You will also need to unroll any loops that you might be inclined to insert into your code.

Q5 **5 points**

Can you formulate any rules of thumb for obtaining good performance when using CUDA? Answer this question in your conclusions.

Part-B: CUDA Unified Memory (20 points)

In this problem we will compare vector operations executed on host vs on GPU to quantify the speed-up.

Q1 4 points

Write a C++ program that adds the elements of two arrays with a K million elements each. Here K is a command-line parameter of the program. Profile and get the time to execute this program for K=1,5,10,50,100. Use `free()` to free the memory at the end of the program.

Q2 **6 points**

Next using CUDA execute the add operation as a kernel on GPU. Use `cudaMalloc()` to allocate memory on GPU for storing the arrays and `cudaMemcpy()` to copy data to and from the GPUs. Note that host and GPU memory is not shared in this case. You will be running three scenarios for this part:

1. Using one block with 1 thread
2. Using one block with 256 threads
3. Using multiple blocks with 256 threads per block with the total number of threads across all the blocks equal to the size of arrays.

Again, profile and get the time to execute this program for $K = 1, 5, 10, 50, 100$. So for each of the three scenarios above you will get profile time for five different values of K Use `cudaFree()` and `free()` to free the memory on the device and host at the end of the program.

Q3 **6 points**

This time you will repeat Step 2 using CUDA Unified Memory and providing a single memory space accessible by all GPUs and CPUs in your system. Instead of `cudaMalloc()` you will use `cudaMallocManaged()` to allocate data in unified memory, which returns a pointer that you can access from host (CPU) code or device (GPU) code. To free the data, just pass the pointer to `cudaFree()`. Again, profile and get the time to execute this program for $K = 1, 5, 10, 50, 100$ for each of the three scenarios.

Q4 **4 points**

Plot two charts one for Step 2 (without Unified Memory) and one for Step 3 (with Unified Memory). The x-axis is the value of K (in million) and y-axis is the time to execute the program (one chart for each of the three scenarios). In both the charts also plot the time to execute on CPU only. Since the execution time scale on CPU and GPU may be orders of magnitude different, you may want to use log-log scale for y-axis when plotting.

For this problem you may want to refer to CUDA slide decks which have a similar example and the following blogpost:
An Even Easier Introduction to CUDA, available at https://developer.nvidia.com/blog/even-easier-introduction-cuda/

Part-C: Convolution in CUDA (and 50 points)

C1: Convolution in CUDA **15 points**

In this part of the lab we implement a convolution of an image using a set of filters. Consider the following:

- An input tensor I with dimensions: C, H, W. Each element of I is generated as follows:

$$I[c, x, y] = c \cdot (x + y)$$

- A set of convolution filters with dimensions: K, C, FH, FW. Each element of the filter F is generated as follows:

$$F[k, c, i, j] = (c + k) \cdot (i + j)$$

- Dimensions are: $H = 1024$, $W = 1024$, $C = 3$, $FW = 3$, $FH = 3$, $K = 64$.
- All tensors have double elements (double precision)
- A tensor I_0 of sizes C, W+2P, H+2P where P=1. I_0 is obtained from the tensor I adding the padding rows and columns with the elements set to zero.
- The output tensor O with dimensions: K, W, H. Each pixel of the output tensor $O[k, x, y]$ is obtained as:

$$O[k, x, y] = \sum_{c=0}^{C-1} \sum_{j=0}^{FH-1} \sum_{i=0}^{FW-1} F[k, c, FW - 1 - i, FH - 1 - j] \cdot I_0[c, x + i, y + j]$$

Note that we need to use the transpose of the filter in order to compute a convolution rather than a cross-correlation. Implement a simple convolution algorithm in CUDA without tiling and without shared memory.

- Print the checksum as the total sum of the elements of O along all its dimensions.
- Report the time to execute the CUDA kernel with the convolution, without including the time to generate the data and to copy it into CUDA memory (only the kernel execution time – you need to use a synchronization primitive).

C2: Tiled Convolution with CUDA 15 points

Implement the convolution at exercise C1 using shared memory and tiling.

- Print the checksum as the sum of the elements of O along all its dimensions. The checksum is expected to be the same as C1.
- Report the time to execute the CUDA kernel with the convolution, without including the time to generate the data and to copy it into CUDA memory (only the kernel execution time – you need to use a thread synchronization primitive).

C3: Convolution with cuDNN 10 points

Implement the convolution at exercise C1 using cuDNN with the *CUDNN_CONVOLUTION_FWD_PREFER_FASTEST* algorithm selection.

- Print the checksum as the sum of the elements of O along all its dimensions. The checksum is expected to be the same as C1.
- Report the time to execute the CUDA kernel with the convolution, without including the time to generate the data and to copy it into CUDA memory (only the kernel execution time).

C4: Convolution with OpenAI Triton (Bonus Question) 10 points

Implement the convolution at exercise C1 using the OpenAI Triton Framework for high-performance deep learning kernels. Please consult the tutorials in this page for usage on Triton: https://triton-lang.org/main/getting-started/tutorials/index.html. At a high level, usage of Triton involves decorating Python functions with `@triton.jit` in order to define a kernel:

```
@triton.jit
def my_kernel(X_ptr, Y_ptr, N):
    # Triton code here
```

Triton operates on memory pointers directly, similar to what we see in CUDA. We have provided the jupyter notebook (`hw3_c4.ipynb`) that includes the skeleton code.

For this problem, please complete two functions (`my_triton_kernel`, `my_conv2d`) and submit the completed jupyter notebook.

Program Output

```
C1_checksum,C1_execution_time
C2_checksum,C2_execution_time
C3_checksum,C3_execution_time
C4_execution_time
```

Where each string contains the checksum (For C1, C2, and C3) and the execution time of the exercises C1, C2, C3 and C4 respectively. Execution times should be printed in milliseconds with 3 decimals. Failing to respect this format is -3 points.

Appendix

Setup for Part-A on Google Cloud Platform

Setting up your GPU instance on GCP

1. Make sure your GPU resource quota supports at least one GPU. To verify check: https://cloud.google.com/compute/resource-usage. To increase the quota, go to GCP Quotas, enable the Compute Engine API under Compute Engine → VM instances, then filter for the metric `compute.googleapis.com/gpus_all_regions`. If your GPU quota is 0, edit it to 1, add a short note, submit, and wait for email confirmation.

2. Create a new instance on GCP with the configuration shown in figure 1. Scroll down to the "Boot disk" option and choose "SWITCH IMAGE". Choose the following image: " Deep Learning VM with CUDA 11.3 M112" (See figure 2). Alternatively you can try this GPU provisioner:

   (a) Download the following files (shared on Drive):

   - `gcp_gpu_provisioner.py` — the provisioning script

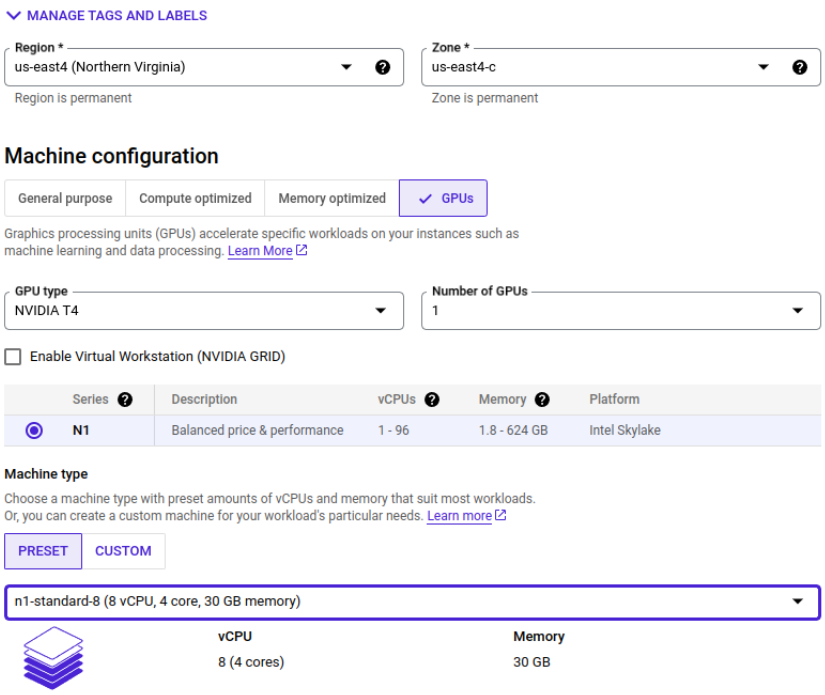

Figure 1: Caption

- `config.json` — configuration template

(b) Authenticate your GCP account:

```
python gcp_gpu_provisioner.py --project PROJECT_ID --authenticate
```

(c) (Optional) View available images:

```
gcloud compute images list --project=ml-images --no-standard-images
```

(d) Edit `config.json` to set:

- Project id
- Disk size, CPU count/type
- GPU type (can start with `nvidia-tesla-t4` for HW3 - easiest to get)

(e) Provision your GPU:

```
python gcp_gpu_provisioner.py --config config.json --provision
```

3. SSH into your instance using the web terminal and when it prompts you to install the NVIDIA CUDA driver, press "y" and the return key. This will automatically install the correct driver on your system. The installer should end with a "Nvidia driver installed" message on your system.

Verifying that Part-A works on your VM instance

1. Upload the 'CUDA1.zip' file to your VM instance using the UPLOAD FILE option at the top in your web terminal.

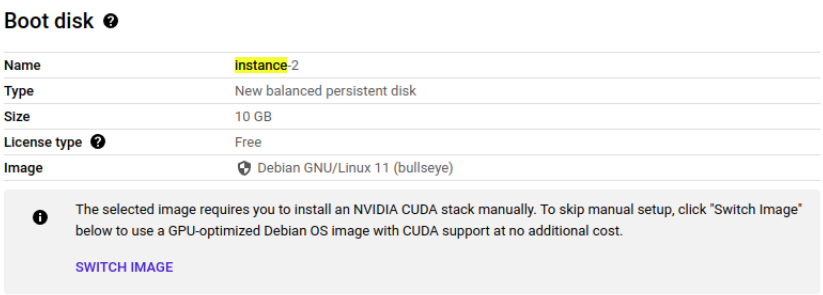

Figure 2: Caption

2. Unzip CUDA1.zip using 'unzip CUDA1.zip' and remove any pre-existing output files using 'rm *.o'.
3. Use 'nano Makefile' or any other editor of your choice. Update the CUDA SDK Install Path (SDK_INSTALL_PATH) to point to your current installation - '/usr/local/cuda'.
4. Now, run 'make' command to build the files.
5. Your installation works correctly if the './vecadd00 500' command gives you 'Test PASSED'.

Frequently Asked Questions

Hardware / Environment

- **Must I run on the course cloud GPU?** No. You can use any GPU system (local or cloud). In your report, list your setup (GPU model, CPU, OS, driver/CUDA versions) so results are interpretable.
- **Finding cloud GPU availability?** Public "GPU finder" utilities may help identify available capacity without repeated manual requests. The GPU provisioner shared above also helps find GPUs.
- **What to do if my installation fails unexpectedly or if I am stuck?** Let one of the TAs know immediately through ED and visit OHs regularly. This homework is particularly long and every single day counts. **Please do not delay letting us know if you are stuck in the setup phase**.

Timing & Measurement

- **What should I time?** Time **kernel execution / kernel invocation**, not full program runtime (setup, allocation, I/O, cleanup). The goal is to compare kernel performance across implementations.
- **Do I need averages?** Not required. You may report a single run or an average over multiple runs. Be consistent across all variants.

- **Why are timings noisy or too similar?** Warm-up effects, caching, and data placement can make results look unexpectedly close or inconsistent. If results look suspicious, rerun with a consistent methodology (same inputs, same warm-up, same timing scope).

CUDA Sync & Version Differences

- **Deprecated CUDA calls?** Using updated equivalents is fine (e.g., newer synchronization APIs), as long as the code synchronizes correctly and timing is captured properly.
- **Where do I synchronize?** Ensure the timer stops **after the kernel finishes** (e.g., sync after launch before reading end time, or use CUDA events correctly).

Warm-up & Memory Experiments

- **Unified vs non-unified look identical after warm-up. Is that bad?** Not necessarily. Warm-up can remove first-run overheads (e.g., first-touch/page migration), making steady-state kernel timings converge. This is acceptable if you explain it.
- **Should I remove warm-up to show a difference?** No need to force differences. Use a consistent benchmarking setup and clearly state whether you measure first-run or steady-state behavior.

Unexpected Performance Results

- **My "optimized" version is slower. Is that possible?** Yes. Small problem sizes, overheads, warm-up/caching, and timing scope can mask benefits.
- **What should I check?** Verify you time kernel-only work, apply the same warm-up strategy across variants, and compare on the same inputs/hardware/software. If still unexpected, document the outcome and methodology clearly.

Reporting

- Include: GPU/CPU, OS, driver/CUDA versions, timing method (kernel-only vs end-to-end), warm-up usage, and whether you report single-run or averaged timings.